RoastIQ — Coffee Cupping Quality & Roasting Line Dashboard

Tool: Microsoft Power BI | Domain: Food Tech / QC / Production Analytics

Project Objective

Design a management dashboard to jointly monitor coffee cupping quality (based on the official Q-Grading system) and roasting/packaging line performance, focused on industry-standard KPIs (Total Cup Points, Specialty Rate, OEE, downtime Pareto).

Data Source

The cupping quality data is modeled on the structure and scoring logic of a well-known, real Kaggle dataset:

- Coffee Quality Database from CQI (kaggle.com/datasets/volpatto/coffee-quality-database-from-cqi) — official Coffee Quality Institute data

This dataset contains genuine scores for Aroma, Flavor, Aftertaste, Acidity, Body, Balance, Uniformity, Clean Cup, Sweetness and Defects for Arabica and Robusta samples from multiple countries. Since this environment has no direct Kaggle download access, values were synthetically generated within the same real ranges and logic; the roasting-line data follows the common structure of Manufacturing Downtime/Efficiency datasets on Kaggle.

Key Performance Indicators (KPIs)

KPI	Formula	Purpose
Total Cup Points	Sum of 10 sensory scores − defect penalty	Final quality score per lot
Specialty Rate	Lots scoring 80+ ÷ total lots	Share of Specialty-grade coffee
OEE	Availability × Performance × Quality	Overall roasting line effectiveness
Category 2 Defects	Average defects per 350g sample	Green bean quality monitoring
Downtime Pareto	Share and cumulative % of each cause	Prioritizing corrective actions

Key Findings

- Arabica averages 80.3 points with a 60% Specialty rate, clearly outperforming Robusta (76.8 average, 3% Specialty rate) — consistent with the global coffee industry standard.
- “Mechanical failure” (30%) and “bag sealing faults” (21%) together account for more than half of total line downtime.
- Line Availability has remained above 90% across all shifts — the line's clearest strength is uptime stability.

Recommended Actions

- Implement a preventive maintenance (PM) program on the equipment with the highest mechanical failure rate.
- Review the bag sealing mechanism to reduce related downtime.
- Optimize the roaster cooling cycle to shorten the gap between roasting batches.

Tools & Method

Power BI Desktop for data modeling, DAX calculations, and visual design; Microsoft Excel for raw data preparation and documentation.

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